



SARDAR PATEL UNIVERSITY
M.Sc.(IT) (SEMESTER - II) EXAMINATION
PS02CINT52: Software Engineering
Thursday, 28th April, 2022

Time: 12:00 pm to 02:00 pm

Max. Marks: 70

- 1.(A) Choose the most appropriate option for each question :** **8**
- (i) UML stands for
(A) Universal Modeling Language (B) Uniform Modeling Logic
(C) Unified Modeling Language (D) Uniform Modeling Language
- (ii) The _____ software development process model provides better risk management and cost for each phase.
(A) Waterfall (B) Spiral (C) Prototyping (D) Iterative
- (iii) The outcome of the requirements analysis phase of the software development life cycle is _____.
(A) a Quality Assurance Plan (B) a Design document (C) an SRS (D) a Test Plan
- (iv) An SRS establishes a basis for agreement between _____.
(A) a user and a product (B) a client and a supplier
(C) a product and a quality (D) developers and a project
- (v) In a use-case diagram, each use case is represented by _____.
(A) an ellipse (B) a circle (C) a rectangle (D) a diamond
- (vi) Which of the following diagrams can show a set of parallel or concurrent flows of activities ?
(A) A class diagram (B) A use-case diagram (C) An activity diagram (D) A flow chart
- (vii) _____ is typically done by the programmer of a module.
(A) Unit testing (B) Beta testing
(C) Black-box testing (D) None of these
- (viii) Which testing is used to validate that a change has not affected some old functionality of the system ?
(A) unit testing (B) regression testing
(C) integration testing (D) white-box testing
- 1.(B) Do as directed.** **16**
Fill in the blanks.
- (i) _____ of a module refers to the relationship among elements of the same module.
- (ii) The _____ model develops software in terms of increments and each increment adds some functionality.
- (iii) _____ is an information system that captures and stores the knowledge of human experts and then imitates human reasoning and decision making process for those who have less expertise.
- (iv) COCOMO stands for _____.
- (v) The _____ notation is used to represent an initial node in the activity diagram.
- (vi) _____ is a process of changing a software system in such a way that it does not alter the external behavior of the code or design, yet improves the internal structure.

C.P. T. O.)

- (vii) In _____ diagrams, the objects participating in the interaction are shown at the top of the chart as boxes attached to a vertical dashed line.
- (viii) _____ testing is performed by "real users" of the software application in "real environment".

State whether the following statements are TRUE or FALSE :

- (ix) A closed system has many interfaces with its environment and it permits interaction across its boundary.
- (x) The spiral model assumes that requirements can be frozen before the design phase begins.
- (xi) A class diagram describes the static structure of a system.
- (xii) In a sequence diagram, an object's lifeline is shown using a vertical dashed line.
- (xiii) White-box testing is also called structural testing.
- (xiv) If two modules communicate using a composite data item, they are stamp coupled.
- (xv) An error refers to the difference between the actual output of a software and the correct output.
- (xvi) The acceptance testing focuses on the external behavior of the system.

2. Answer the following questions (ANY SEVEN):

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- (i) Define : *software engineering*. What are the *goals* of software engineering ?
- (ii) Define the term *system*. Distinguish between closed systems and open systems.
- (iii) Differentiate : *unit testing* vs *integration testing*.
- (iv) What are the *advantages* of a *modular design* ?
- (v) List the *criteria* for determining the *quality* of *software design*.
- (vi) Distinguish between module *coupling* and *cohesion*.
- (vii) What do you mean by *refactoring* ?
- (viii) What is *regression testing* ?
- (ix) What do you mean by *structured programming* ? What is the *advantage* of structured programming ?

3. Describe the *waterfall model* giving its advantages and disadvantages.

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OR

3. Describe the *prototyping model* giving its advantages and disadvantages. How does it differ from the iterative enhancement model ?

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4. Explain the classical *Systems Development Life Cycle (SDLC)* method for developing information systems.

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OR

4. Write a short note on the needs and characteristics of an SRS.

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5. What is UML ? Which *UML diagrams* can be used for *capturing the behavioral view* of a system ? Explain the *class diagrams* giving an example.

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OR

5. Write a short note on *activity diagrams* with an *example*. Provide a summary of various *notations* used in the activity diagrams.

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6. (A) Write a short note on *test oracles*.

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(B) Distinguish between *black-box testing* and *white-box testing*.

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OR

6. Write a short note on *coding standards and coding guidelines*.

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